

UJIAN PRAKTEK BASIS DATA 2



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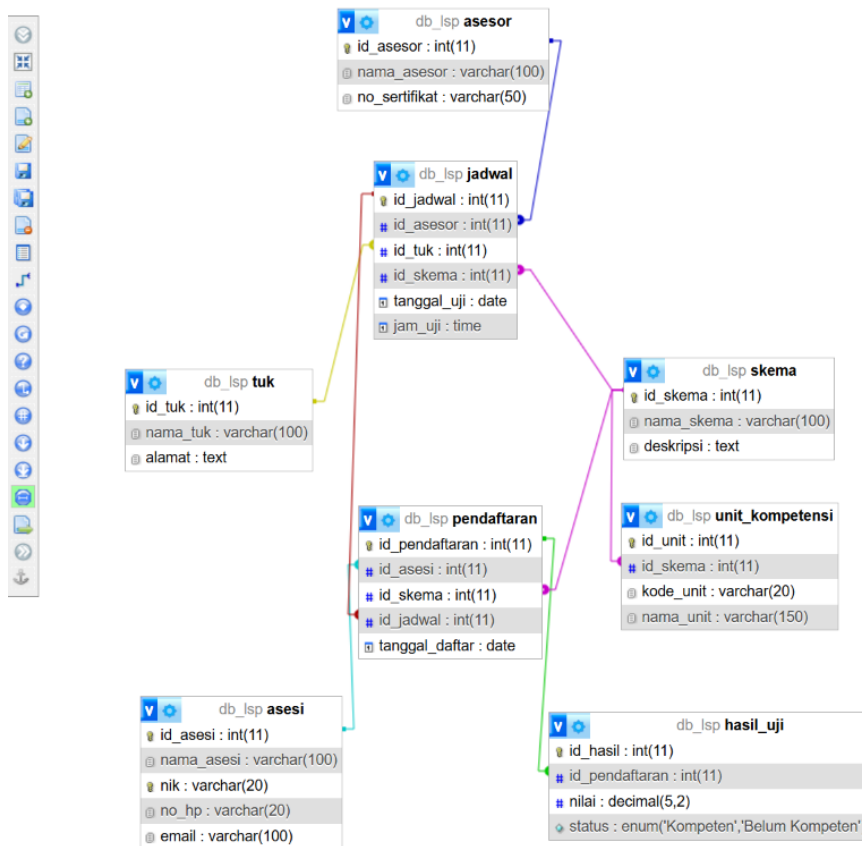
Kelas : 2A

PROGRAM STUDI TEKNIK INFORMATIKA

POLITEKNIK PURBAYA

2026

1. Entity Relationship Diagram (ERD)



2. Perancangan Database

```
MariaDB [(none)]> CREATE DATABASE db_lsp;  
Query OK, 1 row affected (0.011 sec)
```

```
MariaDB [(none)]> USE db_lsp;
```

```
MariaDB [db_lsp]> SHOW DATABASES;
```

```
+-----+  
| Database  
+-----+  
| db_kaba_coffee  
| db_lsp  
| db_siakad  
| db_sidang  
| information_schema  
| minimarket  
| monitoring_jaringan  
| mysql  
| performance_schema  
| phpmyadmin  
| siakad  
| test  
+-----+  
12 rows in set (0.001 sec)
```

3.Membuat tabel

```
Database changed
MariaDB [db_lsp]> CREATE TABLE asesor (
  ->   id_asesor INT AUTO_INCREMENT PRIMARY KEY,
  ->   nama_asesor VARCHAR(100) NOT NULL,
  ->   no_sertifikat VARCHAR(50) NOT NULL
  -> );
Query OK, 0 rows affected (0.087 sec)

MariaDB [db_lsp]> CREATE TABLE asesi (
  ->   id_asesi INT AUTO_INCREMENT PRIMARY KEY,
  ->   nama_asesi VARCHAR(100) NOT NULL,
  ->   nik VARCHAR(20) NOT NULL UNIQUE,
  ->   no_hp VARCHAR(20),
  ->   email VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.125 sec)

MariaDB [db_lsp]> CREATE TABLE skema (
  ->   id_skema INT AUTO_INCREMENT PRIMARY KEY,
  ->   nama_skema VARCHAR(100) NOT NULL,
  ->   deskripsi TEXT
  -> );
Query OK, 0 rows affected (0.021 sec)

MariaDB [db_lsp]> CREATE TABLE tuk (
  ->   id_tuk INT AUTO_INCREMENT PRIMARY KEY,
  ->   nama_tuk VARCHAR(100) NOT NULL,
  ->   alamat TEXT NOT NULL
  -> );
Query OK, 0 rows affected (0.068 sec)

MariaDB [db_lsp]> CREATE TABLE unit_kompetensi (
  ->   id_unit INT AUTO_INCREMENT PRIMARY KEY,
  ->   id_skema INT NOT NULL,
  ->   kode_unit VARCHAR(20) NOT NULL,
  ->   nama_unit VARCHAR(150) NOT NULL,
  ->   FOREIGN KEY (id_skema)
  ->   REFERENCES skema(id_skema)
  -> );
Query OK, 0 rows affected (0.119 sec)
```


4. Data Insert

```
0 rows in set (0.002 sec)

MariaDB [db_lsp]> INSERT INTO asesor (nama_asesor, no_sertifikat) VALUES
-> ('Budi Santoso', 'CERT-001'),
-> ('Siti Aminah', 'CERT-002'),
-> ('Andi Wijaya', 'CERT-003'),
-> ('Dewi Lestari', 'CERT-004'),
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version
MariaDB [db_lsp]> INSERT INTO asesor (nama_asesor, no_sertifikat) VALUES
-> ('Sekar', 'CERT-001'),
-> ('Mega', 'CERT-002'),
-> ('Devi', 'CERT-003'),
-> ('Ismi', 'CERT-004'),
-> ('Tsabita', 'CERT-005');
Query OK, 5 rows affected (0.019 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO asesi (nama_asesi, nik, no_hp, email) VALUES
-> ('Sekar', '3273000000000001', '081234567801', 'sekar@gmail.com'),
-> ('Mega', '3273000000000002', '081234567802', 'mega@gmail.com'),
-> ('Devi', '3273000000000003', '081234567803', 'devi@gmail.com'),
-> ('Ismi', '3273000000000004', '081234567804', 'ismi@gmail.com'),
-> ('Tsabita', '3273000000000005', '081234567805', 'tsabita@gmail.com'),
-> ('Mufida', '3273000000000006', '081234567806', 'mufida@gmail.com'),
-> ('Amel', '3273000000000007', '081234567807', 'amel@gmail.com'),
-> ('Hanin', '3273000000000008', '081234567808', 'hanin@gmail.com'),
-> ('Kirana', '3273000000000009', '081234567809', 'kirana@gmail.com'),
-> ('Lia', '3273000000000010', '081234567810', 'lia@gmail.com'),
-> ('Azam', '3273000000000011', '081234567811', 'azam@gmail.com'),
-> ('Dafin', '3273000000000012', '081234567812', 'dafin@gmail.com'),
-> ('Hanif', '3273000000000013', '081234567813', 'hanif@gmail.com'),
-> ('Sabil', '3273000000000014', '081234567814', 'sabil@gmail.com'),
-> ('Andika', '3273000000000015', '081234567815', 'andika@gmail.com'),
-> ('Fathi', '3273000000000016', '081234567816', 'fathi@gmail.com'),
-> ('Izan', '3273000000000017', '081234567817', 'izan@gmail.com'),
-> ('Ibrahim', '3273000000000018', '081234567818', 'ibrahim@gmail.com'),
-> ('Khirzi', '3273000000000019', '081234567819', 'khirzi@gmail.com'),
-> ('Zahwa', '3273000000000020', '081234567820', 'zahwa@gmail.com');
Query OK, 20 rows affected (0.021 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO skema (nama_skema, deskripsi) VALUES
-> ('Skema Kompetensi Sekar', 'Skema sertifikasi kompetensi bidang teknologi informasi'),
-> ('Skema Kompetensi Mega', 'Skema sertifikasi kompetensi bidang database'),
-> ('Skema Kompetensi Devi', 'Skema sertifikasi kompetensi bidang jaringan'),
-> ('Skema Kompetensi Ismi', 'Skema sertifikasi kompetensi bidang desain grafis'),
-> ('Skema Kompetensi Tsabita', 'Skema sertifikasi kompetensi bidang analisis data');
Query OK, 5 rows affected (0.013 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
C:\WINDOWS\system32\cmd. X + v

MariaDB [db_lsp]> INSERT INTO unit_kompetensi
-> (id_skema, kode_unit, nama_unit) VALUES
-> (1, 'SKR-001', 'Kompetensi Sekar'),
-> (1, 'SKR-002', 'Kompetensi Mega'),
-> (1, 'SKR-003', 'Kompetensi Devi'),
-> (1, 'SKR-004', 'Kompetensi Ismi'),
->
-> (2, 'TSB-001', 'Kompetensi Tsabita'),
-> (2, 'MFD-001', 'Kompetensi Mufida'),
-> (2, 'AML-001', 'Kompetensi Amel'),
-> (2, 'HNN-001', 'Kompetensi Hanin'),
->
-> (3, 'KRN-001', 'Kompetensi Kirana'),
-> (3, 'LIA-001', 'Kompetensi Lia'),
-> (3, 'AZH-001', 'Kompetensi Azam'),
-> (3, 'DFN-001', 'Kompetensi Dafin'),
->
-> (4, 'HNF-001', 'Kompetensi Hanif'),
-> (4, 'SBL-001', 'Kompetensi Sabil'),
-> (4, 'AND-001', 'Kompetensi Andika'),
-> (4, 'FTH-001', 'Kompetensi Fathi'),
->
-> (5, 'IZN-001', 'Kompetensi Izan'),
-> (5, 'IBR-001', 'Kompetensi Ibrahim'),
-> (5, 'KHR-001', 'Kompetensi Khirzi'),
-> (5, 'ZHW-001', 'Kompetensi Zahwa');
Query OK, 20 rows affected (0.020 sec)
Records: 20 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO tuk (nama_tuk, alamat) VALUES
-> ('TUK Sekar', 'Jl. Sekar No. 1'),
-> ('TUK Mega', 'Jl. Mega No. 2'),
-> ('TUK Devi', 'Jl. Devi No. 3'),
-> ('TUK Ismi', 'Jl. Ismi No. 4'),
-> ('TUK Tsabita', 'Jl. Tsabita No. 5');
Query OK, 5 rows affected (0.008 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_lsp]> INSERT INTO jadwal
-> (id_asesor, id_tuk, id_skema, tanggal_uji, jam_uji) VALUES
-> (1, 1, 1, '2026-07-01', '08:00:00'),
-> (2, 2, 1, '2026-07-02', '08:00:00'),
-> (3, 3, 2, '2026-07-03', '09:00:00'),
-> (4, 4, 2, '2026-07-04', '09:00:00'),
-> (1, 5, 3, '2026-07-05', '08:00:00'),
-> (2, 1, 3, '2026-07-06', '09:00:00'),
-> (3, 2, 4, '2026-07-07', '08:00:00'),
-> (4, 3, 4, '2026-07-08', '09:00:00'),
-> (1, 4, 5, '2026-07-09', '08:00:00'),
-> (2, 5, 5, '2026-07-10', '09:00:00');
```

```

MariaDB [db_lsp]> INSERT INTO pendaftaran
-> (id_asesi, id_skema, id_jadwal, tanggal_daftar) VALUES
-> (1, 1, 1, '2026-06-01'),
-> (2, 1, 1, '2026-06-01'),
-> (3, 1, 2, '2026-06-02'),
-> (4, 2, 3, '2026-06-02'),
-> (5, 2, 3, '2026-06-03'),
-> (6, 2, 4, '2026-06-03'),
-> (7, 3, 5, '2026-06-04'),
-> (8, 3, 5, '2026-06-04'),
-> (9, 3, 6, '2026-06-05'),
-> (10, 4, 7, '2026-06-05'),
-> (11, 4, 7, '2026-06-06'),
-> (12, 4, 8, '2026-06-06'),
-> (13, 5, 9, '2026-06-07'),
-> (14, 5, 9, '2026-06-07'),
-> (15, 5, 10, '2026-06-08'),
-> (16, 1, 1, '2026-06-08'),
-> (17, 2, 3, '2026-06-09'),
-> (18, 3, 5, '2026-06-09'),
-> (19, 4, 7, '2026-06-10'),
-> (20, 5, 9, '2026-06-10');
Query OK, 20 rows affected (0.019 sec)
Records: 20 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> INSERT INTO hasil_uji
-> (id_pendaftaran, nilai, status) VALUES
-> (1, 85, 'Kompeten'),
-> (2, 78, 'Kompeten'),
-> (3, 65, 'Belum Kompeten'),
-> (4, 98, 'Kompeten'),
-> (5, 72, 'Kompeten'),
-> (6, 55, 'Belum Kompeten'),
-> (7, 88, 'Kompeten'),
-> (8, 76, 'Kompeten'),
-> (9, 60, 'Belum Kompeten'),
-> (10, 95, 'Kompeten'),
-> (11, 82, 'Kompeten'),
-> (12, 68, 'Belum Kompeten'),
-> (13, 91, 'Kompeten'),
-> (14, 74, 'Kompeten'),
-> (15, 58, 'Belum Kompeten'),
-> (16, 87, 'Kompeten'),
-> (17, 80, 'Kompeten'),
-> (18, 63, 'Belum Kompeten'),
-> (19, 89, 'Kompeten'),
-> (20, 70, 'Kompeten');
Query OK, 20 rows affected (0.018 sec)
Records: 20 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_lsp]> SELECT COUNT(*) AS jumlah_asesi FROM asesi;
+-----+
| jumlah_asesi |
+-----+
|          20 |
+-----+
1 row in set (0.019 sec)

```

```

MariaDB [db_lsp]> SELECT * FROM skema;

```

```

MariaDB [db_lsp]> SELECT * FROM skema;
+----+-----+-----+
| id_skema | nama_skema | deskripsi |
+----+-----+-----+
| 1 | Skema Kompetensi Sekar | Skema sertifikasi kompetensi bidang teknologi informasi |
| 2 | Skema Kompetensi Mega | Skema sertifikasi kompetensi bidang database |
| 3 | Skema Kompetensi Devi | Skema sertifikasi kompetensi bidang jaringan |
| 4 | Skema Kompetensi Ismi | Skema sertifikasi kompetensi bidang desain grafis |
| 5 | Skema Kompetensi Tsabita | Skema sertifikasi kompetensi bidang analisis data |
+----+-----+-----+
5 rows in set (0.001 sec)

```

```

MariaDB [db_lsp]> SELECT COUNT(*) AS jumlah_unit FROM unit_kompetensi;
+-----+
| jumlah_unit |
+-----+
|          20 |
+-----+
1 row in set (0.001 sec)

```

```

MariaDB [db_lsp]> SELECT * FROM tuk;
+----+-----+-----+
| id_tuk | nama_tuk | alamat |
+----+-----+-----+
| 1 | TUK Sekar | Jl. Sekar No. 1 |
| 2 | TUK Mega | Jl. Mega No. 2 |
| 3 | TUK Devi | Jl. Devi No. 3 |
| 4 | TUK Ismi | Jl. Ismi No. 4 |
| 5 | TUK Tsabita | Jl. Tsabita No. 5 |
+----+-----+-----+
5 rows in set (0.001 sec)

```

```
MariaDB [db_lsp]> SELECT COUNT(*) AS jumlah_jadwal FROM jadwal;
+-----+
| jumlah_jadwal |
+-----+
|          10 |
+-----+
1 row in set (0.001 sec)
```

```
MariaDB [db_lsp]> SELECT COUNT(*) AS jumlah_hasil FROM hasil_uji;
+-----+
| jumlah_hasil |
+-----+
|          20 |
+-----+
1 row in set (0.001 sec)
```

```
MariaDB [db_lsp]> SELECT COUNT(*) AS jumlah_pendaftaran FROM pendaftaran;
+-----+
| jumlah_pendaftaran |
+-----+
|          20 |
+-----+
1 row in set (0.001 sec)
```

```
MariaDB [db_lsp]> SELECT 'Asesor' AS tabel, COUNT(*) AS jumlah FROM asesor
-> UNION ALL
-> SELECT 'Asesi', COUNT(*) FROM asesi
-> UNION ALL
-> SELECT 'Skema', COUNT(*) FROM skema
-> UNION ALL
-> SELECT 'Unit Kompetensi', COUNT(*) FROM unit_kompetensi
-> UNION ALL
-> SELECT 'TUK', COUNT(*) FROM tuk
-> UNION ALL
-> SELECT 'Jadwal', COUNT(*) FROM jadwal
-> UNION ALL
-> SELECT 'Pendaftaran', COUNT(*) FROM pendaftaran
-> UNION ALL
-> SELECT 'Hasil Uji', COUNT(*) FROM hasil_uji;
```

tabel	jumlah
Asesor	5
Asesi	20
Skema	5
Unit Kompetensi	20
TUK	5
Jadwal	10
Pendaftaran	20
Hasil Uji	20

```
8 rows in set (0.056 sec)
```

5. View

```
MariaDB [db_lsp]> CREATE VIEW view_sertifikasi AS
-> SELECT
->   a.nama_asesi AS Nama_Asesi,
->   s.nama_skema AS Nama_Skema,
->   ar.nama_asesor AS Nama_Asesor,
->   t.nama_tuk AS Nama_TUK,
->   h.nilai AS Nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p
->   ON h.id_pendaftaran = p.id_pendaftaran
-> JOIN asesi a
->   ON p.id_asesi = a.id_asesi
-> JOIN skema s
->   ON p.id_skema = s.id_skema
-> JOIN jadwal j
->   ON p.id_jadwal = j.id_jadwal
-> JOIN asesor ar
->   ON j.id_asesor = ar.id_asesor
-> JOIN tuk t
->   ON j.id_tuk = t.id_tuk;
Query OK, 0 rows affected (0.056 sec)
```

```
MariaDB [db_lsp]> SELECT * FROM view_sertifikasi;
```

Nama_Asesi	Nama_Skema	Nama_Asesor	Nama_TUK	Nilai
Sekar	Skema Kompetensi Sekar	Sekar	TUK Sekar	85.00
Mega	Skema Kompetensi Sekar	Sekar	TUK Sekar	78.00
Devi	Skema Kompetensi Sekar	Mega	TUK Mega	65.00
Fathi	Skema Kompetensi Sekar	Sekar	TUK Sekar	87.00
Ismi	Skema Kompetensi Mega	Devi	TUK Devi	90.00
Tsabita	Skema Kompetensi Mega	Devi	TUK Devi	72.00
Mufida	Skema Kompetensi Mega	Ismi	TUK Ismi	55.00
Izan	Skema Kompetensi Mega	Devi	TUK Devi	80.00
Anel	Skema Kompetensi Devi	Sekar	TUK Tsabita	88.00
Hanin	Skema Kompetensi Devi	Sekar	TUK Tsabita	76.00
Kirana	Skema Kompetensi Devi	Mega	TUK Sekar	60.00
Ibrahim	Skema Kompetensi Devi	Sekar	TUK Tsabita	63.00
Lia	Skema Kompetensi Ismi	Devi	TUK Mega	95.00
Azam	Skema Kompetensi Ismi	Devi	TUK Mega	82.00
Dafin	Skema Kompetensi Ismi	Ismi	TUK Devi	68.00
Khirzi	Skema Kompetensi Ismi	Devi	TUK Mega	89.00
Hanif	Skema Kompetensi Tsabita	Sekar	TUK Ismi	91.00
Sabil	Skema Kompetensi Tsabita	Sekar	TUK Ismi	74.00
Andika	Skema Kompetensi Tsabita	Mega	TUK Tsabita	58.00
Zahwa	Skema Kompetensi Tsabita	Sekar	TUK Ismi	70.00

```
20 rows in set (0.002 sec)
```

```
MariaDB [db_lsp]> SHOW FULL TABLES;
```

Tables_in_db_lsp	Table_type
asesi	BASE TABLE
asesor	BASE TABLE
hasil_uji	BASE TABLE
jadwal	BASE TABLE
pendaftaran	BASE TABLE
skema	BASE TABLE
tuk	BASE TABLE
unit_kompetensi	BASE TABLE
view_sertifikasi	VIEW

```
9 rows in set (0.013 sec)
```


6. Subquery

```
MariaDB [db_lsp]> SELECT
-> a.nama_asesi,
-> h.nilai
-> FROM asesi a
-> JOIN pendaftaran p
-> ON a.id_asesi = p.id_asesi
-> JOIN hasil_uji h
-> ON p.id_pendaftaran = h.id_pendaftaran
-> WHERE h.nilai > (
-> SELECT AVG(nilai)
-> FROM hasil_uji
-> );
```

nama_asesi	nilai
Sekar	85.00
Mega	78.00
Ismi	90.00
Anel	88.00
Lia	95.00
Azam	82.00
Hanif	91.00
Fathi	87.00
Izan	80.00
Khizrzi	89.00

10 rows in set (0.061 sec)

```
MariaDB [db_lsp]> SELECT
-> ar.id_asesor,
-> ar.nama_asesor
-> FROM asesor ar
-> WHERE ar.id_asesor NOT IN (
-> SELECT j.id_asesor
-> FROM jadwal j
-> );
```

id_asesor	nama_asesor
5	Tsabita

1 row in set (0.008 sec)

```
MariaDB [db_lsp]> SELECT
-> s.nama_skema,
-> COUNT(p.id_pendaftaran) AS jumlah_peserta
-> FROM skema s
-> JOIN pendaftaran p
-> ON s.id_skema = p.id_skema
-> GROUP BY s.id_skema, s.nama_skema
-> HAVING COUNT(p.id_pendaftaran) = (
-> SELECT MAX(jumlah_peserta)
-> FROM (
-> SELECT COUNT(*) AS jumlah_peserta
-> FROM pendaftaran
-> GROUP BY id_skema
-> ) AS jumlah_per_skema
-> );
```

nama_skema	jumlah_peserta
Skema Kompetensi Sekar	4
Skema Kompetensi Mega	4
Skema Kompetensi Devi	4
Skema Kompetensi Ismi	4
Skema Kompetensi Tsabita	4

5 rows in set (0.012 sec)